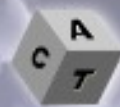


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Enhanced long-term resistance of concrete with marine sessile organisms to chloride ion penetration

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Scientific paper

Enhanced Long-Term Resistance of Concrete with Marine Sessile Organisms to Chloride Ion Penetration

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Abstract

Reinforced concrete (RC) structures in marine environments are generally affected by harsh marine environmental actions, resulting in early performance degradation mainly due to chloride-induced deterioration. In such conditions, corrosion of rebar progresses rapidly, and also the cross-sectional area of rebar is reduced and consequently structural performance of RC structures will be degraded. In contrast, the surface of concrete structures is often covered with many marine sessile organisms under marine tidal and submerged conditions. These marine sessile organisms have been empirically known to enhance the durability of concrete though the effectiveness is not appropriately evaluated.

This paper describes the long-term resistance of concrete with marine sessile organisms to chloride ion penetration in concrete. The effect and its sustainability of marine sessile organisms on chloride ion penetration in concrete were investigated through field exposure tests and laboratory tests. From the test results, the basal membrane, which is a matrix of marine sessile organisms, adheres to concrete strongly on a long-term basis though some gaps between concrete and the basal membrane can be observed. In addition, experimental results and simplified simulation clarified that the attachment of marine sessile organisms can enhance the long-term resistance of concrete to chloride ion penetration.

1. Introduction

Marine reinforced concrete structures are generally affected by harsh marine environmental actions, resulting in early performance degradation mainly due to chloride attack. In such conditions, corrosion of steel bar progresses rapidly, reducing the cross-sectional area of rebar. Consequently, the structural performance of reinforced concrete structures will be degraded.

In the process of chloride-induced corrosion of reinforced concrete, chloride ion penetration into concrete has been discussed all over the world and remains a controversial issue. In order to evaluate the service life of a structure, development of a reliable predictive model of chloride ion penetration in concrete is required. The error function solution to Fick's second law is usually used for practical prediction though many models are proposed in the research field (Nilsson 2009). Chloride ion penetration mechanisms are complex since there are so many factors involved, such as interactions between cement hydrates and chloride, moisture condition of concrete, and time-dependency of diffusion coefficient, to name a few. In addition, the penetration process of concrete immersed in real seawater is different from that immersed in a simulated sodium chloride solution. One

example is interactions between cement hydrates and minor ions in seawater. Buenfeld and Newman (1986) pointed out that the surface skin consisting of a thin layer of brucite of around 30 μm thickness and overlaying thick layer of aragonite can be formed in seawater. This surface skin causes a significant reduction in permeability of concrete. Advanced predictive models considering these phenomena are being developed.

Meanwhile, under marine tidal and submerged environment, the surface of concrete structure is often covered with an abundance of marine sessile organisms such as barnacles, oysters and mussels. It is thought that attachment of these marine sessile organisms is undesired from the aesthetic viewpoint, as is seen by what they are called: "marine fouling organisms." Marine sessile organisms also impede visual inspection of the structure, consequently generating misleading inspection results. Therefore, when carrying out inspection of existing marine structures, the marine sessile organisms is removed from the concrete surface, if necessary (Port and Airport Research Institute 2007).

Regarding the durability of concrete with marine sessile organisms, the previous papers (Iwanami *et al.* 2002, Maruya *et al.* 2003) reported the interesting experimental result that the concrete covered with marine sessile organisms, especially barnacles, had a high resistance to chloride-induced corrosion. The basal membrane of a barnacle is significantly denser than concrete, so the ingress of chloride ion and oxygen can be reduced. This result suggested that there is a possibility that the attachment of marine sessile organisms can contribute to high durability of concrete structures.

The possibility mentioned above poses the question: How long can this effect be maintained? Depending on

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the environmental conditions, the life-span of marine sessile organisms ranges from 1 to 10 years (The sessile organisms society of Japan 2006), apparently shorter than that of concrete structures. In addition, the desiccation process breaks up adhesions between barnacles and attached substances (Wiegemann and Watermann 2004). This point suggests that there is a risk that sufficient effectiveness cannot be attained under the tidal environment for long-term service.

The adhesion between marine sessile organisms and concrete is attributed to the formation of basal membrane consisting of calcite and several types of protein substances. The effect of a cyclic wetting and drying process under the tidal environment condition on adhesions between marine sessile organisms and concrete has not been clarified. Therefore, it is necessary to make clear the long-term resistance of concrete with marine sessile organisms to chloride ion penetration since it can be construed that the adhesion of basal membrane strongly affects the chloride ion penetration.

This paper describes the extensive work of the research reported by previous pioneering works (Iwanami *et al.* 2002, Maruya *et al.* 2003). The long-term exposure testing spanning about 10 years and experiments were carried out in order to determine the long-term resistance of concrete with marine sessile organisms to chloride ion penetration.

2. Details of the experiment

2.1 Test specimens

Reinforced concrete (RC) beams were casted for long-term exposure testing. The size of the test specimens was $200 \times 300 \times 2400$ mm. The mixture proportion of the concrete is summarized in **Table 1**. The water to cement ratio was 49.5 % and ordinary portland cement was used. The compressive strength of the concrete at 37 days was 30.5 N/mm^2 .

The test specimens were subjected to three types of exposure conditions as shown in **Table 2**: (Type A) 9 years and 3 months in Shimizu Port; (Type B) 9 years in the controlled seawater-splashing exposure site in the

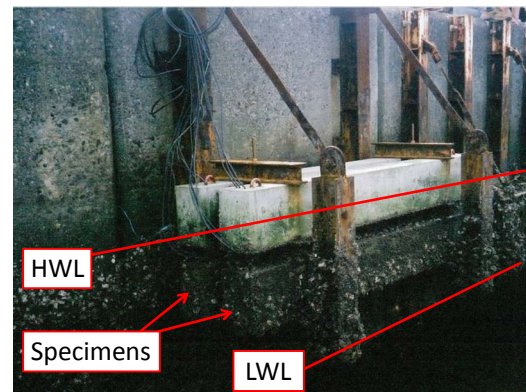


Fig. 1 The location of the test specimens in Shimizu Port.



Fig. 2 Overview of PARI exposure site.

Port and Airport Research Institute (PARI exposure site) after 7 months in Shimizu Port; (Type C) 9 years and 7 months in the PARI exposure site. Shimizu Port is located in inner Suruga Bay. Tidal variation and annual mean temperature were 1.6m and 16.0°C , respectively. According to the investigation of existing RC structure near the exposure site, barnacles, oysters and mussels can be detected as marine sessile organisms. **Figure 1** shows the location of the test specimens in Shimizu Port with the tidal range. The test specimens were immersed in seawater for approximately 15 hours. **Figure 2** shows an overview of the PARI exposure site. At this site, natural seawater was pumped up from Kurihama Bay, which is the inner bay area of Tokyo Bay, and sprayed onto the specimens twice a day for 3 hours each and no marine sessile organisms were attached on the concrete surface. Therefore, the drying period per a day in the PARI exposure site was 18 hours or 2 times longer than that in

Table 1 Mixture proportion of concrete.

W/C (%)	s/a (%)	Unit content (kg/m^3)			
		W	C	S	G
49.5	41.8	161	326	749	1086

Table 2 Exposure condition.

	Exposure condition	Attachment of marine sessile organisms	Attachment area of marine sessile organisms A_{ad} (%)
Type A	9 years and 3 months in Shimizu Port	9 years and 3 months continuously	85-96
Type B	9 years in PARI exposure site after 7 months in Shimizu Port	First 7 months only	42-51
Type C	9 years and 7 months in PARI exposure site	No	0

general tidal environments. The PARI exposure site enhanced the drying process of concrete so that detachment between marine sessile organisms and concrete would be accelerated.

After exposure testing, concrete cores were extracted from the RC beams. By using these core samples, the experimental programs described in the following sections were carried out.

2.2 Attached area measurement

The attached and detached areas of marine sessile organisms were observed and traced on tracing paper. By using image analysis, the percentage of the sum of attached and detached areas to the whole measured area, A_{ad} , was calculated.

2.3 Electrical migration test

Electrical migration testing as an accelerated test is carried out to evaluate the chloride ion penetration in concrete. Cylinder specimens with a diameter of 100 mm were cored from the RC beam. After being saturated with water, the test specimen is sandwiched between two chambers. The specimen which had a thickness of 30 mm, included one surface with marine sessile organisms. As an anodic chamber solution, saturated calcium hydroxide was used, whereas a 5.0 % sodium chloride solution was used as the cathodic chamber solution. The surface with marine sessile organisms was fixed to the cathodic chamber. Direct current of 15 V was applied between the anodic and the cathodic chambers. The sample solution was taken from the anodic chamber periodically to measure the chloride ion concentration. At a steady state, the effective chloride ion diffusion coefficient, D_e , was calculated using the Nernst-Planck's equation with an increasing rate of chloride concentration in the cathodic chamber solution (Japan Society of Civil Engineers 2005).

2.4 Chloride ion content measurement

Chloride ion content in concrete was measured at depths of 0-5, 5-10, 10-20, 20-30, 30-40, 40-50 and 80-90 mm from the surface. After being cut, the samples were powdered by a vibrating mill. Total chloride concentration was measured as per JCI-SC4.

2.5 SEM-EDS analysis

SEM observation and EDS analysis were carried out in order to investigate the interface between marine sessile organisms and concrete. Analyses were conducted with an accelerating voltage of 15 kV and with a beam current of 300 pA.

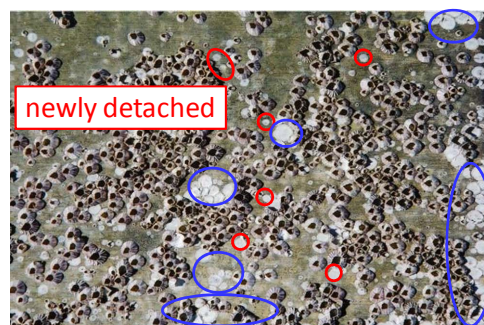
3. Results and discussions

3.1 Attachment of marine sessile organisms

Figure 3 shows the concrete surface of Type B, in which the detachment between marine sessile organisms and concrete surface was accelerated. During exposure in



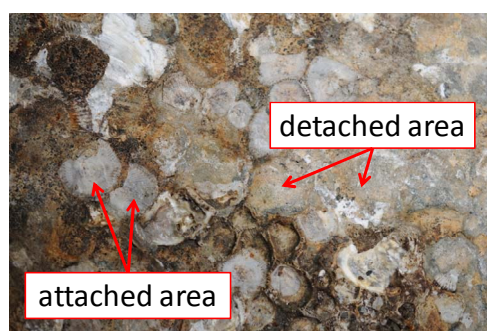
(a) 4 months in PARI exposure site



(b) 7 months in PARI exposure site



(c) Basal membrane



(d) Detachment of basal membrane (after exposure)

Fig. 3 Adhesion of marine sessile organisms to Type B.

Shimizu Port for 7 months, marine sessile organisms were attached on the concrete surface. The main sessile organisms were barnacles and some oysters could be found on the concrete surface. After 4 months of exposure at the PARI exposure site, detachment of barnacles could be found as shown in Fig. 3(a). This might be

caused by transportation from Shimizu Port to the PARI exposure site. Thereafter, newly detachment derived from physical or chemical action could be found as shown in **Fig. 3(b)**. The basal membrane, however, could adhere to the concrete surface (**Fig. 3(c)**). Even after the end of the life-span of marine sessile organisms, the basal membrane can show strong adhesion to attached substances (The sessile organisms society of Japan 2006). After exposure testing, more than half of basal membranes were detached (**Fig. 3(d)**). White gel-like deposits, however, could be detected in the detached area. On the contrary, Type A, in which marine sessile organisms can attach continuously, showed an abundance of marine sessile organisms attached on the concrete surface after exposure. The attached area of marine sessile organisms reached over 85 % of concrete surface. In Type C concrete, no attachment of marine sessile organisms was observed.

In order to investigate adhesion between the basal membrane and concrete, the interface was observed by SEM and analyzed by EDS. **Figure 4** shows the back-scattered electron image (BEI) of Type A concrete. **Figure 4(a)** shows the interface of attached and non-attached areas. Basal membrane adheres to concrete, depending on asperity, voids of the concrete surface. It is also clear that the surface of non-attached area was damaged compared to the attached area. The damage is attributable to seawater attack (Gjorv, 1971). C-S-H gel converting to M-S-H gel and ettringite formation due to penetration of Mg^{2+} and SO_4^{2-} contained in seawater induces damage to the concrete surface. Attachment of

marine sessile organisms can suppress not only chloride ion penetration but also deterioration due to seawater attack. As shown in **Fig. 4(b)**, the basal membrane can also adhere well to the mesoscopically smooth surface of concrete though some gaps between basal membrane and concrete can be detected. **Figure 5** shows a magnified image of the interface between basal membrane and concrete, with distribution of elements. In the microscopic view, physical adhesion is seen between the basal membrane and “microscopic rough” concrete surface but no evidence of chemical adhesion between basal membrane and concrete can be observed. This observation indicates that the adhesion mechanism between basal membrane and concrete is attributed to physical adhesion to the microscopically rough surface.

Figure 4(c) shows the interfaces of attached area and detached area. The gap between basal membrane and concrete and some deposits are observed. Though it is likely shown that there is no basal membrane in the detached area, thin layer can be observed in the magnified image. **Figure 6** shows the magnified image of detached area with distribution of elements. Ca-rich region of which thickness is 20 μm can be observed near the concrete surface. The texture of the region is different from that of cement paste and is dense. This region would be produced by secondary organic-inorganic cement secreted by barnacles. Barnacles, which become partially or wholly detached from their substratum in a natural environment, produce a secondary cement secretion (Saroyan *et al.* 1970). Composition of the secondary secreted cement is similar to that originally secreted

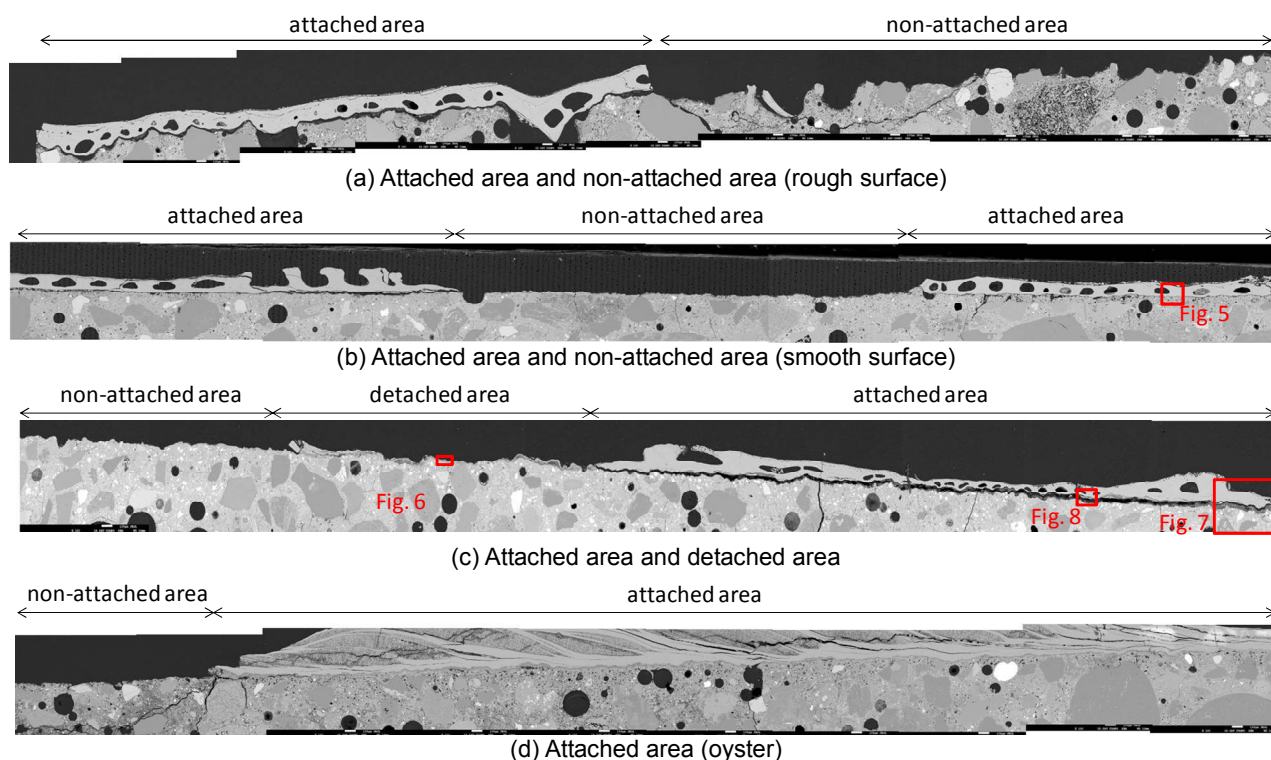


Fig. 4 BEI of interface between concrete and basal membrane (Type A).

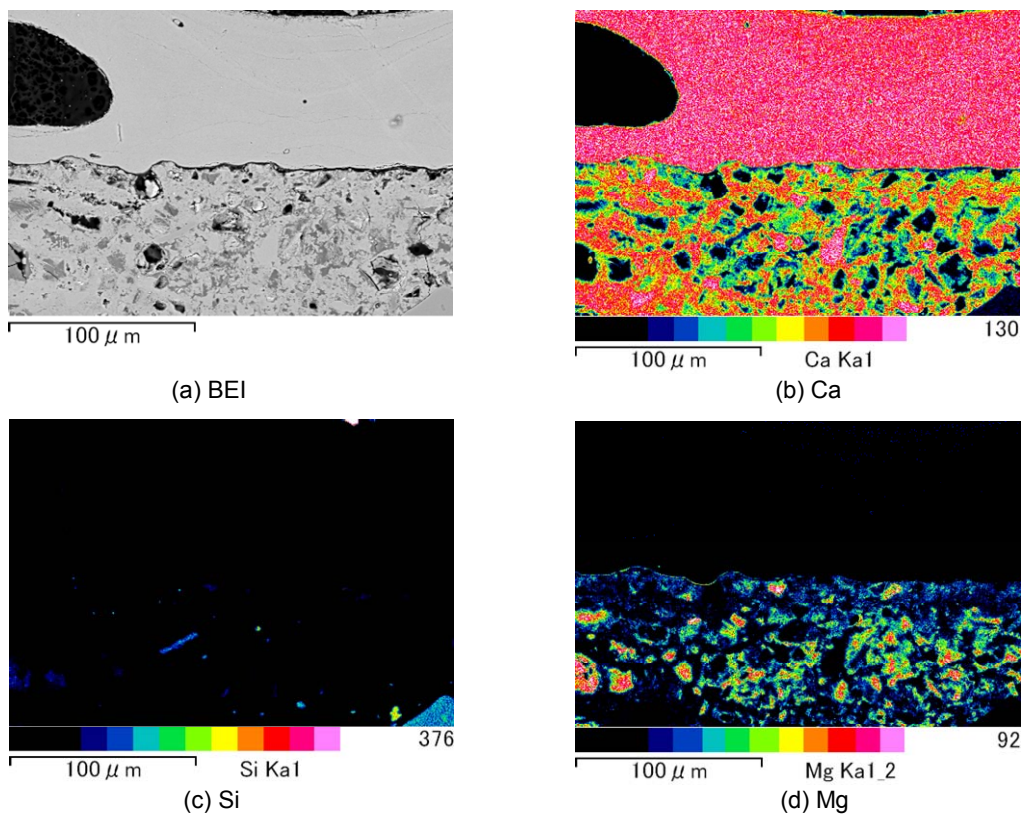


Fig. 5 Magnified image of attached area.

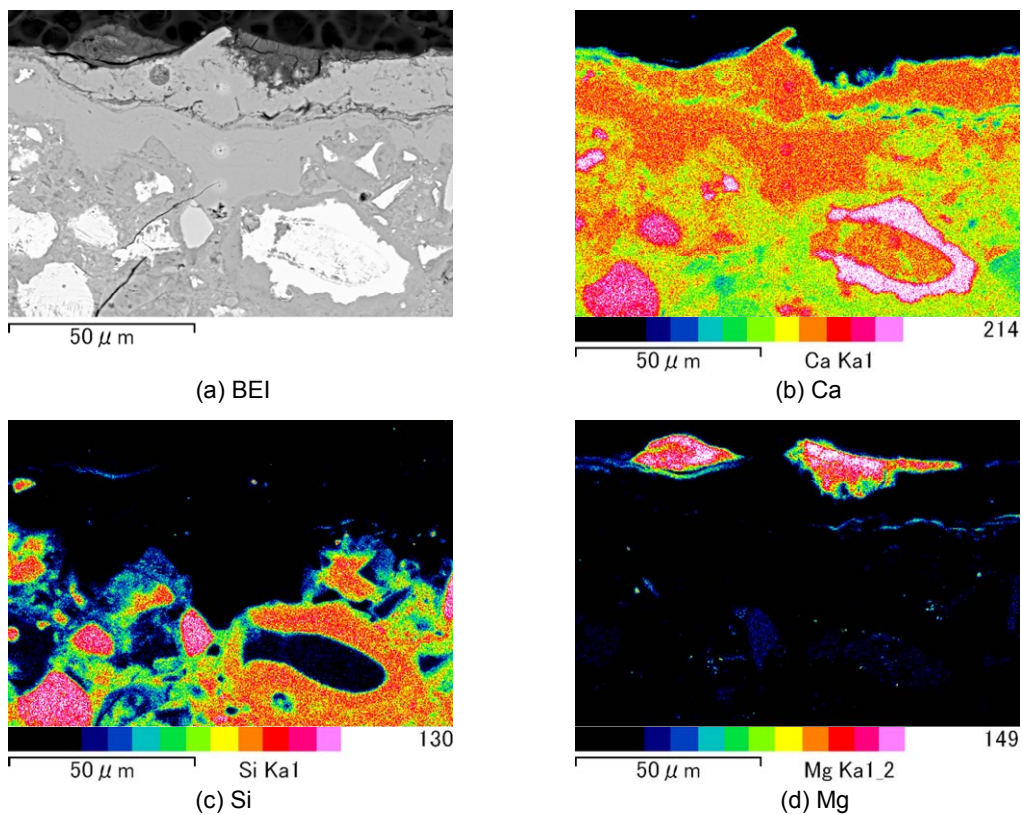


Fig. 6 Magnified image of detached area.

(Kamino *et al.* 1996).

In the gaps, as shown in **Fig. 4(b)** and **Fig. 4(c)**, some deposits are observed. **Figure 7** shows the magnified

image of the boundary between attached and detached areas and distribution of elements. Mg-rich substances, which are supposed to be brucite, are precipitated in the

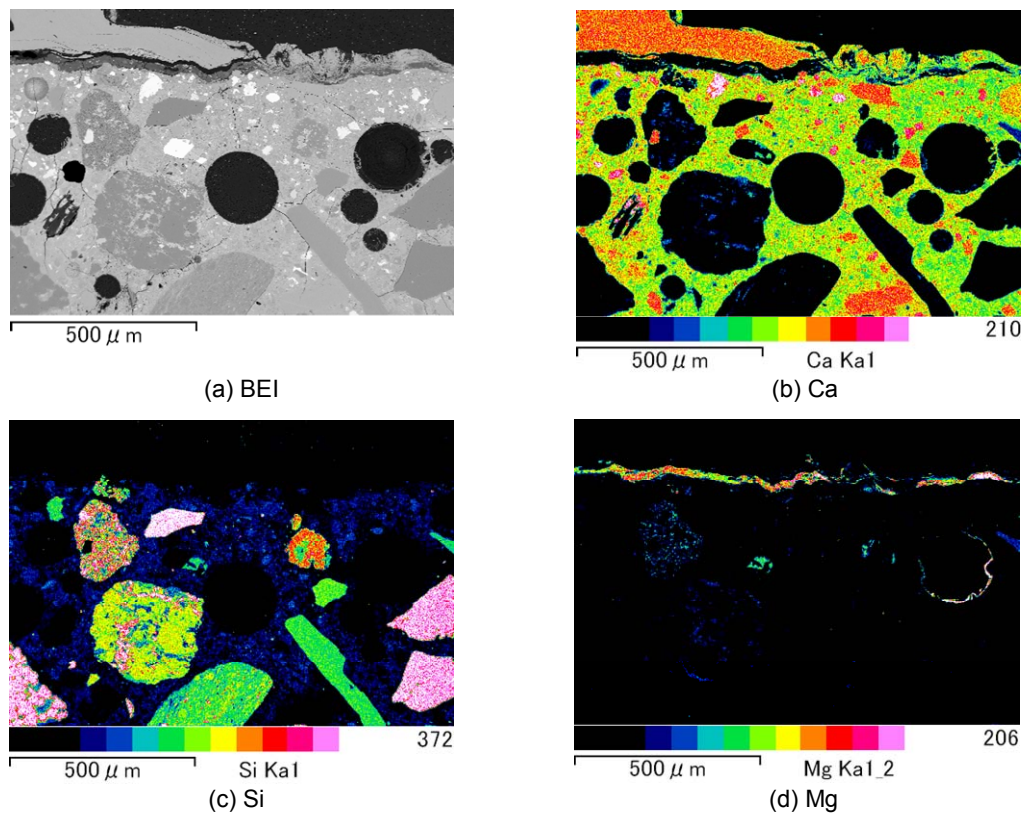


Fig. 7 Magnified image of gap.

gaps. The mechanism is attributed to localized increases in pH within the gap. From the solubility product of brucite, the higher OH^- concentration is, the lower Mg^{2+} . The seawater is saturated with respect to brucite at high pH (Buenfeld and Newman 1986). **Figure 8** shows the magnified image of the gap. It is supposed that the basal membrane originally adheres to concrete well and the gap was not filled with brucite. This means that brucite has nothing to do with gap formation. The precipitation of brucite is not the cause of gap formation, but resulted from gaps derived from other factors.

Figure 4(d) shows the interface between concrete surface and oyster. It is obvious that oyster adheres to the concrete surface compared to barnacles. This observation implies that oysters can also inhibit chloride ion penetration.

3.2 Electrical migration test

Figure 9 shows the effective chloride ion diffusion coefficient, D_e , of concrete with marine sessile organisms. The mean D_e of concrete without marine sessile organisms showed $0.53 \text{ cm}^2/\text{year}$. It should be noted that D_e significantly decreased with an increasing attached area of marine sessile organisms. D_e of concrete with A_{ad} of 96% was only 10% of that without marine sessile organisms. This result suggests that attachment of marine sessile organisms on the concrete surface improves the resistance to chloride ion penetration.

In the experiment carried out in the previous research (Iwanami *et al.* 2002, Maruya *et al.* 2003), D_e of 6 types

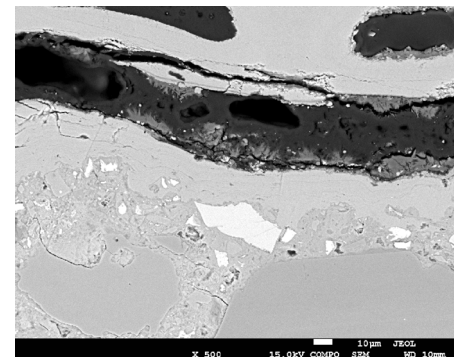
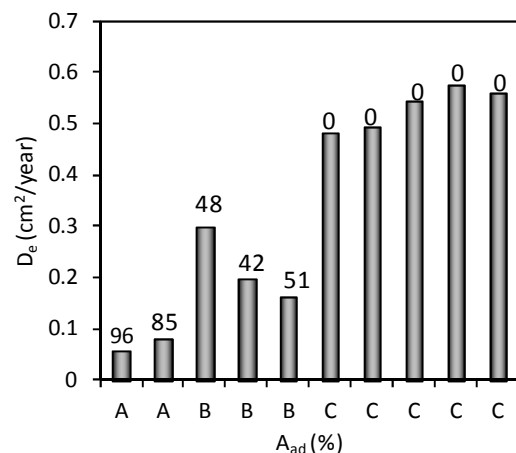


Fig. 8 Magnified image of gap.

Fig. 9 D_e of concrete with/ without marine sessile organisms.

of concrete with different materials and mixture proportions was evaluated. The experimental conditions are the same as that described in this paper except for the shape of the test specimen. The result showed that the attached area of marine sessile organisms strongly affects D_e while the material properties and mixture proportions demonstrate no significant effect. **Figure 10** shows the relationship between A_{ad} , attached and detached areas of marine sessile organisms, and the normalized D_e of the concrete. The normalized D_e is defined as the ratio of the D_e of concrete to that of non-attached concrete. For comparison, the data published in the previous paper are plotted in this figure. In general, after about a 10-year exposure, the trend between A_{ad} and the normalized D_e is almost the same as that published in the previous paper, irrespective of exposure conditions. Type B, in which the detachment between marine sessile organisms and concrete surface was accelerated, also showed the consistent trend. In fact, about more than half of the marine sessile organisms were detached from the concrete surface in Type B. It is interesting that the sum of the attached area and detached area is correlated to normalized D_e . This result suggests that not only the attached area but also the detached area inhibit chloride ion penetration on a long-term basis.

According to Wiegmann and Watermann (2004), barnacles cannot adhere to non-stick coatings when being desiccated. The drying process of a tidal zone, however, allows marine sessile organisms to be supplied with seawater or moisture from the concrete. The supply of water from concrete would help to keep the adhesion between marine sessile organisms and concrete surface. From these results, long-term resistance of concrete to chloride ion penetration can be promising in a natural marine tidal zone when an abundance of marine sessile organisms are attached on the concrete surface.

3.3 Chloride ion content

The chloride content profiles in concrete Type A with a different attached areas are shown in **Fig. 11**. The chloride content decreased with increase in the distance from the concrete surface. **Figure 12** shows the total chloride ion content in concrete. The total chloride content penetrated into concrete with marine sessile organisms was less than half that of non-attached concrete. This result strongly supports that the attachment of marine sessile organisms prevents chloride ion from penetrating into concrete. In the tidal zone and submerged zone where the marine sessile organisms can attach continuously, high resistance of concrete to chloride ion penetration can be attained.

In order to obtain the apparent diffusion coefficient of concrete, the error function solution to Fick's second law was fitted to the chloride content profile in the concrete without marine sessile organisms, as shown in Eq. (1).

$$C(x, t) = C_0 \left(1 - \operatorname{erf} \left(\frac{x}{2\sqrt{D_a t}} \right) \right) \quad (1)$$

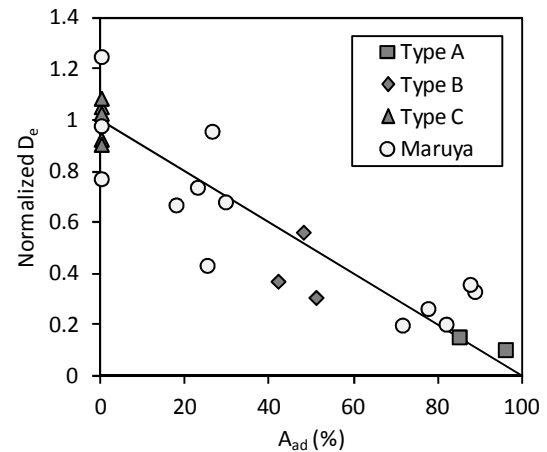


Fig. 10 A_{ad} and normalized D_e .

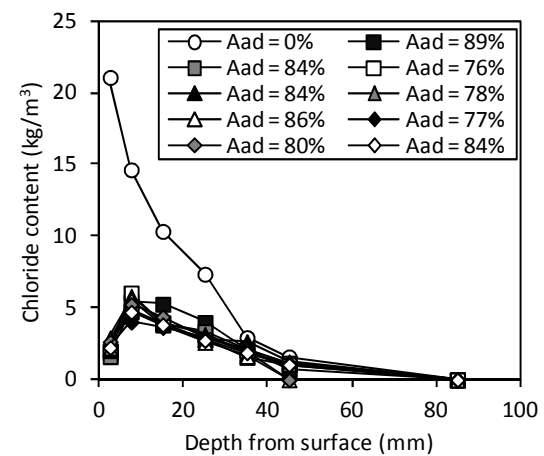


Fig. 11 Chloride content profiles of Type A concrete.

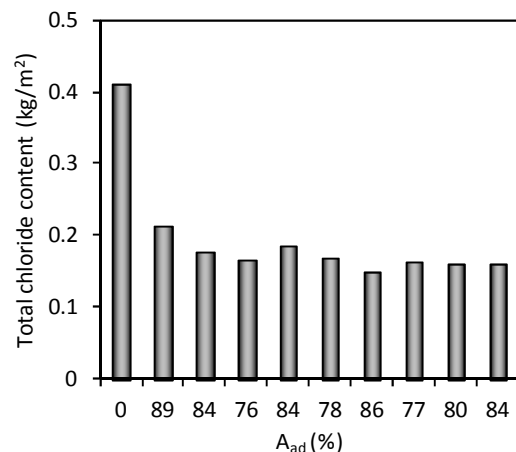


Fig. 12 Total chloride content in Type A concrete.

where, $C(x, t)$ = chloride content at depth x and time t , C_0 = chloride content at surface, x = depth from concrete surface, t = time, and D_a = apparent diffusion coefficient. From the calculation, the apparent diffusion coefficient of concrete and chloride content at surface was 0.29

cm^2/year , 21.5 kg/m^3 , respectively. This calculation does not take into consideration time-dependent changes in either apparent diffusion coefficient or chloride content at surface.

Here, the effect of the marine sessile organisms on chloride ion diffusivity in concrete was considered. In the actual condition, marine sessile organisms attach continuously so that chloride ion penetration would decrease with the increase in attached area simultaneously. Simply assumed that the diffusion coefficient of basal membrane, D_m , is zero, chloride ion cannot penetrate from the attached area. This means the area where the chloride ion ingresses decreases with increase in attached area. Time-dependent change of attachment of marine sessile organisms is difficult to evaluate quantitatively as was attempted in the previous research (Maruya *et al.* 2003). For simplified evaluation, the time evolution of attachment of marine sessile organisms is expressed as in the following Eqs. (2) and (3).

$$A_{ad(t)} = A_{ad\max} \times U_{(t)} \quad (2)$$

$$U_{(t)} = \begin{cases} 1(t \geq t_a) \\ 0(t < t_a) \end{cases} \quad (3)$$

where, $A_{ad(t)}$ = the sum of attached and detached areas at time t , $A_{ad\max}$ = maximum value of A_{ad} , $U_{(t)}$ = step function, t_a = the period at which $A_{(t)}$ equals to $A_{ad\max}$, and t = time. According to Maruya *et al.* (2003), A_{ad} reached the maximum value at the exposure period of 1 or 2 years so that t_a was set to 0.5, 1.5 and 2 years. From the experiments, $A_{ad\max}$ was set to be 80%. Before t equals to t_a , the chloride ion diffusion was calculated according to Eq. (1). After t_a , the diffusion at attached and detached areas and non-attached area are calculated individually. In the attached and detached areas, the calculation method for surface coated concrete was used (Tsutsumi *et al.* 2010). Thereafter, the chloride content profiles are calculated with weighted average method. For simplified calculation, one dimensional diffusion was assumed. **Table 3** summarizes the calculation condition.

Figure 13 shows the calculated results. Calculated chloride profiles show similar trends to experimental results. The difference between calculation and experiment at the concrete surface would be attributed to carbonation and Ca-leaching from concrete. It is also clear that the effect of t_a on chloride profiles was small in this calculation. More importantly, the calculated results support that chloride ion had not been able to penetrate into concrete after 1 to 2 years of exposure. Assuming that such high resistance of concrete to chloride ion penetration will be sustained permanently, chloride ion content at 70 mm from the concrete surface, which is one of standard covering depths, can be calculated as 1.2 kg/m^3 after 50-year service.

Chloride content profiles and calculations greatly suggest that the attachment of marine sessile organisms can enhance the long-term resistance of concrete to chloride ion penetration. In the previous paper, the ap-

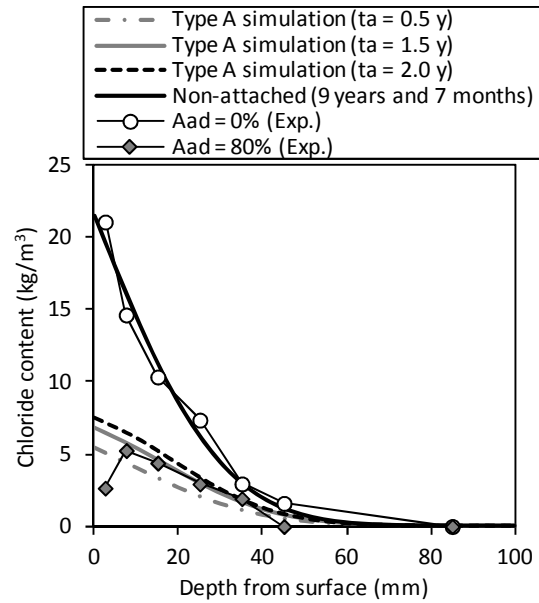


Fig. 13 Calculated chloride content profiles.

Table 3 Calculation condition.

C_0 (kg/m^3)	21.5
D_a (cm^2/year)	0.29
D_m (cm^2/year)	0
t_a (year)	0.5, 1.5, 2
$A_{ad\max}$ (%)	80

parent diffusion coefficient of concrete covered with marine sessile organisms is 100 times smaller than that of non-attached concrete. In the previous paper, however, the apparent diffusion coefficient was decided by exposure testing for 1 year and 8 months. It is well known that the apparent diffusion coefficient is a time-dependent parameter so the diffusion coefficient of marine sessile organisms would be a significantly smaller value, nearly equal to zero.

3.4 Effect of attachment of marine sessile organisms on chloride ion penetration

From the results described above, the attachment of marine sessile organisms can enhance the long-term resistance of concrete structures to chloride ion penetration. Even if the marine sessile organisms are detached from the concrete surface, the secondary secreted cemented region would prevent the penetration of chloride ion. In existing marine structures, deterioration of concrete structures located in tidal and submerged zones is rarely more severe than our expectations. The result is well consistent with the empirical rule.

4. Conclusions

This paper investigated the long-term resistance of concrete with marine sessile organisms to chloride ion penetration. The exposure test spanned about 10 years and experiments were carried out. The conclusions can

be summarized as follows:

- (1) The basal membrane, which is a dense matrix of marine sessile organisms, adheres to concrete strongly on a long-term basis though some gaps between concrete and the basal membrane can be observed.
- (2) In the detached area of the basal membrane, the region containing a high Ca content can be observed. This region would be produced by a secondary organic-inorganic cement secretion.
- (3) The electrical migration test revealed that the sum of attached and detached areas of the basal membrane is well correlated to effective diffusion coefficient of concrete, irrespective of exposure duration. This result suggests that not only the attached area but also the detached area of marine sessile organisms can inhibit chloride ion penetration on a long-term basis.
- (4) Total chloride ion content in concrete with marine sessile organisms was significantly less than non-attached concrete. Experimental results and simplified simulation clarified that the attachment of marine sessile organisms can enhance the long-term resistance of concrete to chloride ion penetration.

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